

When the knowledge in a script is used to guide agent action, the agent must create an instance of it, which is called a plan. A plan differs from a script in that (a) it selects a particular path through the often-variable sequence of events permitted by the script and (b) it fills the events' case roles in particular ways.

The events and objects referred to in a script are, themselves, concepts that are recorded in appropriate branches of the ontology. This means that scripts are organized like well-constructed computer programs—not as a massive main function but, rather, as hierarchically organized drilldowns of scripts that ultimately end in singleton events or function calls.

We will describe scripts using the example of the AGENT-FUNCTIONING-FLOW script that implements the agent's cognitive architecture and makes the agent self-aware so that it can explain its own functioning (cf. section 2.1). This script uses the EVENTS and OBJECTS shown in the ontological subtrees in table 3.2, which we provide as a crib to be consulted when reading the scripts themselves. This description of AGENT-FUNCTIONING-FLOW serves the double duty of presenting an example of a script as an ontological entity and describing an important aspect of LEIA operation.

The top level of the AGENT-FUNCTIONING-FLOW script has five ordered subevents that correlate with the architecture diagram in figure 2.1.²¹

AGENT-FUNCTIONING-FLOW

DEFINITION This script implements the LEIA's cognitive architecture.

AGENT LEIA-1

SUBEVENTS

- 1. PERCEPTION-RECOGNITION
- 2. PERCEPTION-INTERPRETATION
- 3. DELIBERATION
- 4. ACTION-SPECIFICATION
- 5. ACTION-RENDERING

All of the subevents are, themselves, scripts, which we will describe in turn. The first one is PERCEPTION-RECOGNITION, whose function is described in its definition field.

PERCEPTION-RECOGNITION

DEFINITION The agent determines which recognizer is needed to process a signal, runs it, and stores the resulting data.

AGENT LEIA

INPUT PERCEPTION-INPUT

OUTPUT PERCEIVED-DATA

3. The action can be triggered by a daemon (a standing goal). For example, a particular virtual patient might need to know if procedures are painful before agreeing to them. If the doctor recommends a procedure for which the patient lacks information about pain, it posts the goal of tracking down that information.
4. The action can be the next step in the current plan on the agent's agenda, launched when the previous step is completed. For example, if a robotic LEIA is building a chair, then after it attaches the first leg, it undertakes to attach the second leg, and then the third, and the fourth.

In all cases, the agent knows why it chose a given action. Formally, this is recorded as metadata with the remembered instance of the event. For cases 2–4, explaining the reasoning is relatively straightforward, apart from some details of language generation. If the agent is asked *Why did you say that?* or *Why are you doing that?* it answers:

- 2a. Because [a brief description of the first event in the adjacency pair]. For example, Q: *Why did you say that?* A: *Because Joe asked me where I was going.*
- 3a. Because [a brief description of the daemon]. For example, Q: *Why did you ask that question?* A: *Because I need to know if procedures are painful before agreeing to them.*
- 4a. It's the next step in [plan name]. For example, Q: *Why are you doing that?* A: *It's the next step in building a chair.*

By contrast, explaining decision functions, the first case above, is more complicated since those decisions can involve not only agent actions but also recommendations, advice, warnings, and so on. Examples of what a person might want to know about decisions include, among many others:

- Why did you do that?
- Why did you recommend that?
- Why do you think that what I was planning to do is wrong?
- What did you base your recommendation on?
- Was your recommendation informed by machine learning?
- Would any additional information be useful in making this decision?
- How sure are you of this recommendation?
- Are you sure this is best?
- Why is [option X] better than [option Y]?